



Real-Time Facial Emotion Recognition with Meme Display

Course: BS in Data Science and Artificial Intelligence

Institute: Indian Institute of Management Sambalpur

1. Objective

The objective of this project is to design and implement a real-time facial emotion recognition system using a Convolutional Neural Network (CNN). The system aims to identify human emotions from live webcam input and display a corresponding meme image. This project demonstrates the application of data science, artificial intelligence, and computer vision techniques in a practical and interactive manner.

2. Process Followed

The project was implemented following a structured approach:

- **Dataset Selection:** The FER-2013 dataset was used, which consists of grayscale facial images categorized into seven emotion classes.
- **Data Preprocessing:** Images were resized to 48×48 pixels and normalized to improve model performance.
- **Model Development:** A CNN model was developed using TensorFlow and Keras to learn facial features and classify emotions.
- **Model Training:** The CNN was trained on the dataset and saved as a trained model file.
- **Face Detection:** OpenCV Haar Cascade was used to detect faces from live webcam input.
- **Emotion Prediction:** The detected face was passed to the trained CNN to predict emotions.
- **Meme Display:** A predefined meme corresponding to the detected emotion was displayed.
- **Visualization:** A live emotion frequency chart was generated to show emotion trends.

3. Results

The system successfully detects facial emotions in real time from a live webcam feed. Detected emotions are displayed accurately along with the corresponding meme images. The emotion frequency chart updates dynamically, providing a visual summary of detected emotions over time. The system performs efficiently and provides stable output without frequent flickering.



4. Conclusion

This project demonstrates the effective use of Convolutional Neural Networks and computer vision techniques for real-time facial emotion recognition. The integration of meme display adds an interactive element, making the application engaging and user-friendly. The project successfully meets its objectives and serves as a strong academic demonstration of AI and data science concepts.

5. Future Work

The project can be enhanced in several ways:

- Improving model accuracy using advanced CNN architectures or transfer learning.
- Adding a graphical user interface (GUI) for better user interaction.
- Supporting multiple face detection simultaneously.
- Storing emotion data for long-term analysis and reporting.
- Integrating emotion-based recommendations or analytics.

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